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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1

Programme: B.E-CSE/B.Tech-IT
Course Code: SCSB1202
Course Name: DATA STRUCTURES
Batch : 2025-2026

Max. Marks: 50
Time: 2 Hours
Sem: II
Date: 16.02.2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Define Data Structure.	CO1
2.	Define array and list its types.	CO1
3.	Tabulate the difference between Array and Linked List.	CO1
4.	Draw the node structure of a Doubly Linked List.	CO1
5.	List the operations performed on a STACK.	CO2
6.	Explain the need of Priority Queues.	CO2
7.	Explain Circular Queue with a neat diagram.	CO2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Illustrate in detail the classification of data structures with examples and list the operations applied to linear data structure.	CO1
	(OR)	
9.	Describe the algorithm to perform the insertion of an element at the beginning of an ARRAY. Discuss the application of the algorithm using a suitable example.	CO1
10.	Enumerate the operations performed on a LINKED LIST. Discuss the algorithm to perform deletion of a node at the end of the linked list with a suitable example.	CO1
	(OR)	
11.	Define Stack. Outline the algorithm for Pop() operation on a STACK data structure with an example.	CO2
12.	Covert the given infix expression to a postfix expression: ((A+B)-C*(D/E))+F. Write the algorithm for infix to postfix conversion.	CO2
	(OR)	
13.	Define Queue. Mention the algorithm for Enqueue() operation and explain it with an example.	CO2